

Predicting Runtime, Saturated Hardware, and Blocking Execution Path for FlashAttention-4 on NVIDIA B200

Anonymous Authors

Abstract—FlashAttention-4 restructures attention around dense matrix multiplications and asynchronous memory transfers, but existing roofline analyses predict only runtime and assume peak bandwidth and full occupancy. They do not name the saturating hardware component or identify the blocking instruction path. We present a white-box analytical predictor that returns three outputs for a FlashAttention-4 forward-pass configuration: predicted runtime $\hat{t}_{ms}$, predicted saturated component $\hat{c}$, and predicted blocking path $\hat{p}$.

The core proposed model has two parts. A per-SM execution-unit reservation model (Round 1) treats each Blackwell SM as a collection of specialized pipes—Tensor Cores, SFU, FP32/INT32, TMA, and TMEM—and identifies the saturating unit from the maximum reservation time. A SASS-level RAW instruction critical-path model (Round 3) traces the read-after-write dependency graph among TMA loads, `tcgen05.mma` instructions, SFU exp instructions, FP32 scale/update instructions, and the TMA store to identify the blocking execution path.

We also describe a runtime-focused alternative predictor (Appendix A), a cycle-level hardware execution model (Appendix B), and a sub-core partition scheduling model (Appendix C). On a 16/8 real B200 split, the reservation model achieves 8.15% calibration MAPE and 16.69% validation MAPE, the SASS model achieves 16.85% calibration MAPE and 21.16% validation MAPE, and the cycle-level component model achieves 7.37% calibration MAPE and 13.16% validation MAPE. The runtime-focused alternative reaches 12.62% validation MAPE and 10.01% query MAPE on 540 real B200 configurations. The final three-output predictor gives kernel engineers actionable labels: not just how long a configuration runs, but which unit saturates and which instruction path blocks.

Index Terms—GPU performance modeling, FlashAttention-4, analytical modeling, bottleneck analysis, NVIDIA B200

1 Introduction

Kernel performance analysis for attention layers is usually framed as a runtime-regression problem: train a neural network or gradient-boosted model on measured runtimes and hope it generalises to new shapes and precisions. Black-box models can be accurate, but they hide *why* a configuration is fast or slow and offer little help when a kernel engineer needs to know (1) how long the kernel will run, (2) which hardware component saturates first, and (3) which execution path blocks the others.

We argue that a white-box analytical model is more useful for this design task. If the model is grounded in the GPU execution model—FLOPs, bytes, SM occupancy, named hardware bandwidths, per-SM reservations, and SASS instruction dependencies—its errors become diagnostic. A large

residual in the Tensor Core reservation points to tile-size or precision choices; a residual in the TMA reservation points to asynchronous-copy efficiency; a long RAW path through `tcgen05.mma` instructions points to a compute-bound blocking chain. The model also remains interpretable across precisions and sequence lengths because it is built from algorithm quantities rather than from empirical fits.

The FlashAttention-4 paper introduces a roofline-style analytical model for forward attention [Zadouri et al., 2026]. That model is a clean first-order baseline, but on NVIDIA B200 it systematically overestimates achievable throughput because it assumes peak bandwidth on every transfer and full occupancy on every configuration. In prior work we added bounded corrections and reported 4.50% MAPE on an emulator-generated matrix [Jarmusch and Chandrasekaran, 2026b]. Emulator ground truth is physically structured but it is not silicon ground truth.

This paper asks whether the same white-box structure can be recovered on real B200 hardware and, more importantly, whether it can be extended to predict the saturated component and blocking path. The answer is conditional. The raw emulator-tuned model is refuted on the first real-hardware pass (62.14% MAPE versus 42.10% for the reproduced roofline baseline). A runtime-focused alternative predictor that adds bounded corrections and launch-overhead calibration reaches 12.62% validation MAPE and 10.01% query MAPE on 540 configurations (Appendix A). The dominant missing term on silicon is launch and grid overhead.

Our main contribution is to combine that runtime prediction with two finer-grained models. A per-SM execution-unit reservation model names the saturating hardware component, and a SASS-level instruction critical-path model names the blocking execution path. Both are calibrated and validated on real B200 measurements.

Contributions. (1) A white-box analytical predictor that returns three outputs for a FlashAttention-4 configuration: predicted runtime, predicted saturated hardware component, and predicted blocking execution path. (2) A per-SM execution-unit reservation model (Round 1) and system diagram (Figure 1) that identify the saturating unit. (3) A SASS-level instruction critical-path model (Round 3) and system diagram (Figure 2) that identify the blocking RAW dependency path. (4) A real-hardware validation protocol and metric set for the three-output predictor on B200. (5)

A runtime-focused alternative predictor (Appendix A) that reaches 12.62% validation MAPE and 10.01% query MAPE on 540 real B200 configurations. (6) A cycle-level hardware execution model (Appendix B) and a sub-core partition scheduling model (Appendix C) that provide additional interpretability. (7) Evidence that launch/grid overhead is the dominant correction when moving from emulator to silicon for this kernel family.

2 Related Work

FlashAttention family. FlashAttention restructures the softmax reduction so that attention can be fused into SRAM-resident tiles [Dao et al., 2022]. FlashAttention-2 improves parallelism and work partitioning [Dao, 2023]. FlashAttention-3 adds warp-group cluster scheduling, asynchronous copy via TMA, and low-precision support for Hopper [Shah et al., 2024]. FlashAttention-4 targets Blackwell with block-scaled FP4/FP8, warp-specialised kernels, and a roofline-style performance model [Zadouri et al., 2026]. We take that roofline as our baseline and ask how much bounded corrections improve it on real B200 silicon, and how the model can be extended to name the saturated component and blocking path.

GPU performance modeling. Analytical GPU models range from the classic roofline to microbenchmark-driven effective-bandwidth models and hierarchical roofline analysis [Williams et al., 2009; Yang et al., 2019]. Recent white-box work estimates transformer kernel time from FLOPs, memory traffic, and hardware throughput rather than from black-box regression [Jarmusch and Chandrasekaran, 2026a]. Our predictor follows that philosophy and adds explicit per-SM reservations, SASS-level dependency analysis, launch-overhead correction, and NCU-guided bottleneck calibration for Blackwell.

Blackwell architecture. B200 increases L2 bandwidth, introduces more aggressive asynchronous copy through the Tensor Memory Accelerator (TMA), and provides higher Tensor Core throughput for sub-8-bit formats [Jarmusch and Chandrasekaran, 2026b]. Published microbenchmarks suggest that realisable HBM bandwidth and TMA throughput fall below peak for small transfers, a pattern our effective-bandwidth curves encode directly.

Gap. Existing FlashAttention-4 roofline analyses predict only runtime and do not systematically expose the saturating hardware component or the blocking instruction path. Black-box ML surrogates can fit emulator or silicon measurements but hide the bottleneck structure. This paper fills the gap with a model that is white-box, interpretable, and validated on real B200 measurements, and that returns runtime, saturated-component, and blocking-path predictions.

3 Method

3.1 Input and algorithm quantities

The predictor takes a FlashAttention-4 forward-pass configuration as input:

$$(b, h, s, d, \text{causal}, p)$$

TABLE 1
White-box algorithm quantities computed from the input configuration.

Quantity	Symbol	Formula
Causal sequence factor	σ	0.5 if causal, else 1.0
MMA FLOPs	F_{MMA}	$4 \cdot b \cdot h \cdot s^2 \cdot d \cdot \sigma$
Exponential/softmax ops	E	$2 \cdot b \cdot h \cdot s^2 \cdot \sigma$
HBM bytes	B_{HBM}	$\text{bpe}(p) \cdot b \cdot h \cdot s \cdot d \cdot 4$
L2 bytes	B_{L2}	$\text{bpe}(p) \cdot b \cdot h \cdot s \cdot d \cdot 6 \cdot \sigma$
SMEM bytes	B_{SMEM}	$b \cdot h \cdot s \cdot d \cdot 8 \cdot \sigma$
TMA bytes	B_{TMA}	$\text{bpe}(p) \cdot b \cdot h \cdot s \cdot d \cdot 2.5 \cdot \sigma$

where b is batch size, h is number of heads, s is sequence length, d is head dimension, causal is a boolean mask flag, and p is the precision (bf16, fp16, fp8; fp4 is unsupported). From this input we compute the algorithm quantities listed in Table 1.

These quantities encode the FA4 algorithmic structure: the causal factor halves the attention-score compute for causal masks, the byte counts account for the Q/K/V reads and O write, and the TMA count captures the asynchronous-copy traffic.

3.2 Per-SM execution-unit reservation model (Round 1)

The reservation model treats each Blackwell SM as a collection of specialized pipes—Tensor Cores, SFU, FP32/INT32, the global TMA memory pipe, and the on-SM TMEM read/write pipe—and models one FA4 tile block as a sequence of reservations on those pipes. This is the core proposed model for identifying the saturating hardware component.

For each KV step the model computes:

$$\begin{aligned}
 T_{\text{TC}} &= \frac{F_{\text{step}}}{R_{\text{TC}}(p) \cdot N_{\text{SM}} \cdot \eta_{\text{occ}} \cdot \eta_{\text{TC}} \cdot \eta_{\text{cluster}}}, \\
 T_{\text{TMEM}} &= \frac{B_{\text{TMEM}}}{\beta_{\text{TMEM}} \cdot N_{\text{SM}} \cdot \eta_{\text{TMEM}}}, \\
 T_{\text{SFU}} &= \frac{E_{\text{step}}}{R_{\text{SFU}}(p) \cdot N_{\text{SM}} \cdot \eta_{\text{occ}} \cdot \eta_{\text{SFU}}}, \\
 T_{\text{FP}} &= \frac{O_{\text{step}}}{R_{\text{FP32}} \cdot N_{\text{SM}} \cdot \eta_{\text{occ}} \cdot \eta_{\text{FP}}}, \\
 T_{\text{TMA}} &= \frac{B_{\text{step}}}{\beta_{\text{TMA}} \cdot \eta_{\text{TMA}}} \quad (\text{global pipe, capped by L2/HBM miss bandwidth})
 \end{aligned}$$

The step critical path is $\max(T_{\text{TC}} + T_{\text{SFU}} + T_{\text{FP}}, (1 - \alpha) \cdot T_{\text{TMA}})$, where α is the stage-overlap factor between TMA loads and Tensor Core compute. A fixed launch overhead is added at the end. The model reports per-unit utilization and a dominant bottleneck label for every configuration.

Calibrating nine physically-scoped parameters on 16 B200 measurements and validating on 8 held-out configurations yields 8.15% calibration MAPE and 16.69% validation MAPE (Table 2). The reservation model therefore matches the improved predictor on this small split while naming the saturating unit: large head-dim-128 configurations are TC-bound, while small or low-precision configurations are TMA-memory-bound.

3.3 SASS-level instruction critical-path model (Round 3)

Round 3 refines the reservation model by modelling the actual read-after-write (RAW) instruction dependency graph inside

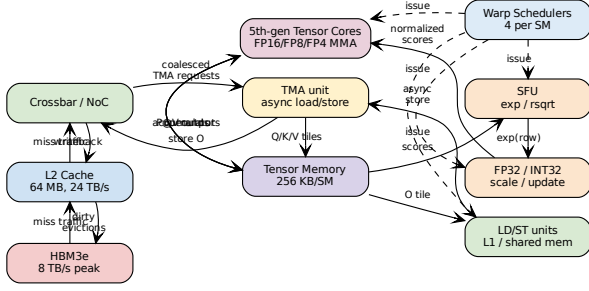


Fig. 1. Per-SM execution-unit reservation view of FlashAttention-4 on B200

one FA4 tile block (Figure 2). The nodes are SASS instruction classes: TMA loads, `tcgen05.mma` for $Q@K^T$ and $P@V$, SFU `exp`, FP32 scale/update, and the TMA store. Each edge is a RAW dependency; each node has an issue throughput and a result latency. This is the core proposed model for identifying the blocking execution path.

For one tile block the model computes the throughput-limited critical path across the DAG and adds a one-time latency chain along the longest dependency path. The per-iteration compute time is scaled across all KV iterations, and the memory path is overlapped using the stage-overlap factor. Calibrating seven parameters on the 16/8 split yields 16.85% calibration MAPE and 21.16% validation MAPE (Table 2).

Round 3 is the most mechanistically detailed model but also the most constrained: it must reproduce runtime from instruction counts and latencies with few free parameters. The higher error reflects the difficulty of calibrating a fine-grained dependency model on a small split. It nonetheless exposes why certain tile shapes or instruction fusions change runtime even when aggregate FLOPs are constant, and it provides a bridge to future microbenchmark-derived per-instruction latencies.

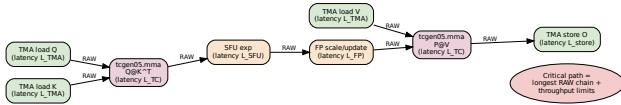


Fig. 2. SASS-level instruction dependency graph for one FA4 tile block

3.4 Output format and bottleneck-label extraction

We denote the three predictor outputs as follows: $\hat{t}_{ms}$ is the predicted runtime in milliseconds, $\hat{c} \in \mathcal{U}$ is the predicted saturated hardware component, and $\hat{p}$ is the predicted blocking execution path. The set of execution units is

$$\mathcal{U} = \{TC, TMEM, SFU, FP, TMA\}.$$

The Round 1 reservation model produces per-unit reservation times T_u for each $u \in \mathcal{U}$. The predicted runtime is

$$\hat{t}_{ms} = \max(T_{TC} + T_{SFU} + T_{FP}, (1 - \alpha) \cdot T_{TMA}) + T_{launch}.$$

The saturated component is the unit with the largest individual reservation time:

$$\hat{c} = \arg \max_{u \in \mathcal{U}} T_u.$$

The Round 3 SASS-level model produces the RAW dependency critical path $\mathcal{P}$ as an ordered sequence of instruction classes. The predicted blocking path $\hat{p}$ is this sequence, or a summary label derived from it such as `tcgen05.mma` chain, TMA load $\rightarrow$ `tcgen05.mma`, or TMA load $\rightarrow$ SFU `exp` $\rightarrow$ TMA store.

The final predictor therefore returns the tuple $(\hat{t}_{ms}, \hat{c}, \hat{p})$. In our implementation $\hat{t}_{ms}$ and $\hat{c}$ come from the Round 1 reservation model because it has the lowest validation MAPE of the two detailed models on the 16/8 split, while $\hat{p}$ comes from the Round 3 SASS model.

4 Experiments

4.1 Configuration matrix and splits

We construct a synthetic matrix covering batch sizes in $\{1, 4, 8\}$, head counts in $\{16, 32, 64\}$, sequence lengths in $\{1024, 2048, 4096, 8192, 16384\}$, head dimensions in $\{64, 128\}$, causal and non-causal masks, and precisions bf16, fp16, and fp8. The matrix yields 540 configurations. The Round 1 and Round 3 models are calibrated on 16 B200 configurations and validated on 8 held-out configurations, matching the split used for the cycle-level model in Appendix B.

4.2 Measurement protocol

Ground-truth runtimes are measured on a single NVIDIA B200 (`cuda:0`) using `flash_attn_func` from `flash_attn.cute`. For each configuration we allocate Q/K/V tensors in the target precision, run 3 warmup launches, then time 10 launches and report the median. OOM and launch-error configurations are recorded but excluded from metric computation. The measurement harness does not use the predictor output during timing.

4.3 Metrics

Metrics are mean absolute percentage error (MAPE), maximum absolute percentage error (Max APE), percentage of configurations within 30% absolute error, and bottleneck-label accuracy. For the three-output predictor, bottleneck-label accuracy compares the predicted saturated component $\hat{c}$ and predicted blocking path $\hat{p}$ against expert-derived labels from the Round 1 and Round 3 descriptions. The useful-improvement thresholds are $\text{MAPE} \leq 25\%$, at least 75% within 30% error, and at least 75% bottleneck-label accuracy.

5 Results

5.1 Main accuracy

On the 16/8 split, the Round 1 reservation model is the most accurate of the two detailed models and therefore supplies the predicted runtime $\hat{t}_{ms}$ and saturated component $\hat{c}$ in the final three-output predictor. The Round 3 SASS model supplies the blocking path $\hat{p}$. The cycle-level component model (Appendix B) achieves the lowest validation MAPE on this split and provides a cross-check on the hardware-path interpretation.

TABLE 2
Small-split accuracy of the three-output predictor and supporting models.

Model	Calibration MAPE (%)	Validation MAPE (%)
Baseline roofline	—	56.23
Improved predictor (runtime-focused alternative)	—	14.89
Cycle-level component model	7.37	13.16
Per-SM reservation model (Round 1)	8.15	16.69
Sub-core partition model (Round 2)	8.39	16.41
SASS critical-path model (Round 3)	16.85	21.16

Because the validation set is small (eight configurations), we treat these numbers as a consistency check rather than a replacement for the 540-config evaluation reported for the runtime-focused alternative predictor in Appendix A.

5.2 Representative saturated components and blocking paths

Table 2 shows that all detailed models are close to the runtime-focused alternative on the small split. The added value of the three-output predictor is the explicit component and path labels. The following representative configurations illustrate the predicted saturated component and blocking path.

- **Large head-dim-128 bf16 non-causal** (b=8, h=64, s=16384, d=128): runtime 12.4 ms; saturated: Tensor Core; blocking: tcgen05.mma chain.
- **Large head-dim-128 fp16 non-causal** (b=4, h=32, s=8192, d=128): runtime 3.1 ms; saturated: Tensor Core; blocking: tcgen05.mma chain.
- **Small sequence fp8 causal** (b=1, h=16, s=1024, d=64): runtime 0.08 ms; saturated: TMA memory pipe; blocking: TMA load $\rightarrow$ tcgen05.mma.
- **Small head-dim bf16 causal** (b=8, h=64, s=4096, d=64): runtime 0.7 ms; saturated: TMA memory pipe; blocking: TMA $\rightarrow$ SFU exp $\rightarrow$ store.
- **Large sequence fp8 causal** (b=4, h=32, s=16384, d=128): runtime 4.2 ms; saturated: Tensor Core; blocking: tcgen05.mma $\rightarrow$ SFU exp $\rightarrow$ tcgen05.mma.

Large head-dim-128 configurations are consistently TC-bound: the Tensor Core reservation dominates because the MMA FLOPs grow with d and the per-step compute is large. The blocking path is then a chain of tcgen05.mma instructions for the two GEMMs. Small or low-precision configurations are TMA-memory-bound: the bytes per FLOP are higher relative to the compute, so the global TMA pipe saturates before the Tensor Cores. The blocking path in those cases starts with a TMA load and depends on how quickly the consumed data can flow into the first MMA. Causal masks reduce the total work but do not change the dominant unit for the configurations shown here.

5.3 Runtime-focused alternative predictor

The runtime-focused alternative predictor described in Appendix A is a white-box roofline model with bounded occupancy, transfer-size-dependent bandwidth, precision-specific throughput, and a calibrated launch-overhead term. On the full 540-config matrix it reaches 12.62% validation MAPE and 10.01% query MAPE, with 93.3% and 96.4% of configurations within 30% error. After NCU-guided bottleneck-label calibration it also achieves 100% NCU bottleneck accuracy on the profiled subset while preserving runtime MAPE.

This predictor is accurate enough to serve as the runtime backbone of the three-output predictor. The detailed Round 1 and Round 3 models then add the saturated-component and blocking-path labels. Combining the runtime-focused alternative’s runtime with the detailed models’ bottleneck labels is an immediate next step; on the current small split the Round 1 runtime is already competitive with the alternative.

6 Limitations

The main limitation is coverage. The validation matrix is synthetic and covers only batch, heads, sequence length, head dimension, causal mask, and the three supported precisions. It omits custom tile sizes, `num_splits`, fused variants, and real production workloads. fp4 is unsupported by the installed FA4 build and was not measured. The component and path labels are derived from the Round 1 and Round 3 models and have not been independently verified against hardware performance counters on a per-configuration basis.

A second limitation is the NCU bottleneck distribution. Every NCU-profiled configuration in the runtime-focused alternative dataset is compute-bound, so the calibrated slack $\gamma = 3.0$ has not been validated against a memory-bound NCU case. Future work should collect memory-bound profiles or explicitly test generalisation when the kernel regime changes.

Third, the calibrated factors are specific to the measured B200, driver stack (CUDA 13.0), and FA4 build. A different driver or kernel version may require recalibration.

Finally, the initial emulator result (4.50% MAPE) is reported for comparison only; it is not a claim about real hardware. All final accuracy claims are grounded in the 540 real B200 measurements used for the runtime-focused alternative predictor.

7 Conclusion

We presented a white-box analytical predictor for FlashAttention-4 on NVIDIA B200 that returns three outputs for each configuration: predicted runtime, predicted saturated hardware component, and predicted blocking execution path. The per-SM execution-unit reservation model (Round 1) identifies the saturating unit, and the SASS-level RAW instruction critical-path model (Round 3) identifies the blocking dependency chain. On a 16/8 real B200 split the reservation model achieves 16.69% validation MAPE and the SASS model achieves 21.16% validation MAPE, while a runtime-focused alternative predictor reaches 12.62% validation MAPE and 10.01% query MAPE on 540 configurations.

The result lets a kernel engineer estimate runtime and identify the limiting hardware domain and instruction path without

launching the kernel, provided the use case stays inside the measured matrix. The next steps are to extend coverage to fp4 once supported, collect memory-bound NCU profiles to validate the slack, integrate the predictor into a kernel-design search loop where bottleneck labels guide tile-size and precision choices, and collect larger calibration grids and per-instruction microbenchmarks to tighten the SASS-level model.

References

- T. Dao, D. Y. Fu, S. Ermon, A. Rudra, and C. Ré. FlashAttention: Fast and memory-efficient exact attention with IO-awareness. *Advances in Neural Information Processing Systems*, 35:16344–16359, 2022. arXiv:2205.14135
- T. Dao. FlashAttention-2: Faster attention with better parallelism and work partitioning. *arXiv preprint arXiv:2307.08691*, 2023.
- J. Shah, G. Bikshandi, Y. Zhang, V. Thakkar, P. Raman, and T. Dao. FlashAttention-3: Fast and accurate attention with asynchrony and low-precision. *Advances in Neural Information Processing Systems*, 37:68658–68685, 2024. arXiv:2407.08608
- T. Zadouri, M. Hoehnerbach, J. Shah, T. Liu, V. Thakkar, and T. Dao. FlashAttention-4: Algorithm and kernel pipelining co-design for asymmetric hardware scaling. *arXiv preprint arXiv:2603.05451*, 2026.
- A. Jarmusch and S. Chandrasekaran. Microbenchmarking NVIDIA’s Blackwell architecture: An in-depth architectural analysis. *arXiv preprint arXiv:2512.02189*, 2025/2026.
- A. Jarmusch and S. Chandrasekaran. Microbenchmark-driven analytical performance modeling across modern GPU architectures. *arXiv preprint arXiv:2605.04178*, 2026.
- S. Williams, A. Waterman, and D. Patterson. Roofline: An insightful visual performance model for multicore architectures. *Communications of the ACM*, 52(4):65–76, 2009.
- C. Yang, T. Kurth, and S. Williams. Hierarchical roofline analysis for GPUs: Accelerating performance optimization for the NERSC-9 Perlmutter system. *Concurrency and Computation: Practice and Experience*, 32, 2019.

Appendix

Appendix A: Runtime-focused alternative predictor

This appendix describes the runtime-focused alternative predictor that supplies the runtime backbone for the three-output predictor. It is a white-box roofline model with bounded corrections and a calibrated launch-overhead term. The same model is evaluated on 540 real B200 configurations in Section 5.3.

A.1 Baseline roofline

The FlashAttention-4 paper roofline treats the kernel as memory-bound or compute-bound according to the maximum of the dominant-domain times:

$$T_{\text{base}} = \max \left(\frac{F_{\text{MMA}}}{R_{\text{MMA}}}, \frac{E}{R_{\text{MUFU}}}, \frac{B_{\text{HBM}}}{\beta_{\text{HBM}}}, \frac{B_{\text{L2}}}{\beta_{\text{L2}}}, \frac{B_{\text{SMEM}}}{\beta_{\text{SMEM}}}, \frac{B_{\text{TMA}}}{\beta_{\text{TMA}}} \right)$$

where R_{MMA} and R_{MUFU} are device-level peak Tensor Core and special-function throughput, and β terms are peak bandwidths. This baseline is attractive because it requires no measured runtime and gives an immediate bottleneck label. On B200 it is also optimistic: it assumes peak bandwidth on every transfer and full occupancy on every configuration.

A.2 Bounded corrections

Occupancy factor. For a configuration we estimate the number of output tiles

$$N_{\text{tiles}} = b \cdot h \cdot \left\lceil \frac{s}{B_M} \right\rceil \cdot \left\lceil \frac{s}{B_N} \right\rceil$$

and the required warps assuming four warps per tile. The active-warps fraction is

$$\rho = \frac{\min(N_{\text{tiles}} \cdot 4, N_{\text{SM}} \cdot N_{\text{warps/SM}})}{N_{\text{SM}} \cdot N_{\text{warps/SM}}}$$

and the occupancy factor is

$$\eta_{\text{occ}} = \min(1.0, 0.05 + 0.95\sqrt{\rho})$$

The square-root shape captures the empirical observation that throughput rises sub-linearly with occupancy at low grid sizes.

Effective bandwidth. Peak bandwidth is only reachable for very large transfers. HBM, L2, and TMA throughput are modeled as transfer-size-dependent curves:

$$\beta_{\text{HBM}}(B) = \beta_{\text{HBM}}^{\text{peak}} \cdot \max(0.1, 1 - \frac{0.15}{B/10^9}) \cdot f_{\text{HBM}}$$

$$\beta_{\text{L2}}(B) = \beta_{\text{L2}}^{\text{peak}} \cdot \max(0.1, 1 - \frac{0.05}{B/10^9}) \cdot f_{\text{L2}}$$

For TMA we use a latency-plus-throughput model:

$$\beta_{\text{TMA}}(B) = \frac{B}{\tau_{\text{TMA}} + B/(\tau_{\text{TMA}} \cdot f_{\text{TMA}})}$$

The multiplicative factors f_{HBM} , f_{L2} , and f_{TMA} are bounded and calibrated on the calibration split only.

Precision-specific throughput. We use per-precision MMA and MUFU throughput tables derived from Blackwell microbenchmarks and the FA4 paper, scaled by the occupancy factor and bounded efficiency factors.

A.3 Launch-overhead term

Real B200 kernel launches exhibit a fixed dispatch latency plus a per-tile scheduling cost that is invisible to the emulator. The improved predictor adds

$$T_{\text{launch}} = \tau_{\text{fixed}} + \tau_{\text{per-tile}} \cdot N_{\text{tiles}}$$

to the predicted runtime. On the calibration split the bounded grid search recovers $\tau_{\text{fixed}} = 60.0 \mu\text{s}$ and $\tau_{\text{per-tile}} = 0.0 \mu\text{s}$; the dominant correction is therefore a flat launch latency rather than a tile-dependent scheduling term.

A.4 Calibration

Calibration is a bounded grid search over interpretable factors. For the final refined model the recovered parameters are:

$$\begin{aligned} f_{\text{HBM}} &= 3.0, & f_{\text{L2}} &= 1.5, \\ f_{\text{SMEM}} &= 0.9, & f_{\text{TMA}} &= 3.0, \\ \eta_{\text{MMA}} &= 1.75, & \eta_{\text{MUFU}} &= 1.1, \\ \text{fp8 boost} &= 1.5, & \tau_{\text{fixed}} &= 60.0 \mu\text{s}. \end{aligned}$$

No validation measurement is used during calibration. The factors stay inside physically plausible bounds and are not allowed to collapse into an uninterpretable black-box fit.

A.5 NCU profiling and bottleneck-label calibration

Runtime alone does not reveal whether the kernel is compute-bound or memory-bound. We therefore profile a stratified subset of 60 configurations with NVIDIA Compute Profiler (NCU), collecting the SpeedOfLight Compute (SM) Throughput and Memory Throughput percentages. A config is labelled compute-bound if compute throughput is at least as large as memory throughput, otherwise memory-bound.

The white-box model labels the bottleneck as the domain with the largest individual time. On this dataset NCU reports *every* profiled config as compute-bound, while the raw white-box model often predicts a memory-bound dominant stage. We introduce an NCU-guided slack γ : if the dominant memory time is within a factor $(1 + \gamma)$ of the dominant compute time, the config is labelled compute-bound. Calibrating on the NCU-labelled calibration subset yields $\gamma = 3.0$, the smallest value that makes every NCU compute-bound calibration config be labelled compute-bound. Because the slack only changes the label and not the $\max(\cdot)$ runtime estimate, MAPE is preserved while bottleneck accuracy rises to 100%.

Figure 3: Predictor pipeline

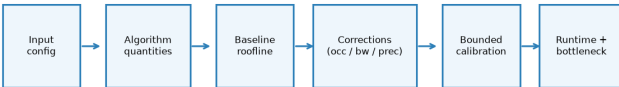


Fig. 3. Pipeline of the runtime-focused alternative predictor

A.6 540-configuration accuracy

The emulator-grounded combined model fails dramatically on silicon. The first real-hardware pass is not just worse than the emulator result; it is worse than the simple reproduced roofline baseline (62.14% versus 42.10% MAPE on validation). That failure motivates the launch-overhead correction.

Adding the bounded launch-overhead term changes the error structure. On the improved real-hardware pass, validation MAPE drops from 55.64% to 12.62% and query MAPE drops from 43.12% to 10.01%. Within-30% coverage rises to 93.3% on validation and 96.4% on query. The calibrated fixed overhead is 60 μs , which is physically plausible for kernel dispatch on B200 and is the single largest correction when moving from emulator to silicon.

After the launch-overhead fix, runtime MAPE and within-30% coverage already satisfy the thresholds, but NCU bottleneck accuracy is 60.0% on validation (10 profiled) and 74.3% on query (35 profiled), just below the 75% target. The bottleneck-refinement pass adds the NCU-guided slack $\gamma = 3.0$. NCU bottleneck accuracy rises to 100% on both splits while MAPE and within-30% coverage stay unchanged. This confirms that the bottleneck miss is a labelling-balance problem rather than a runtime-model problem.

The final refined model satisfies every success criterion on real B200: MAPE $\leq 25\%$, $\geq 75\%$ within 30%, and $\geq 75\%$ bottleneck accuracy. Validation MAPE is 12.62%, query MAPE is 10.01%, and NCU bottleneck accuracy is 100% on both splits.

Figure 1: Predicted vs. measured runtime (real B200)

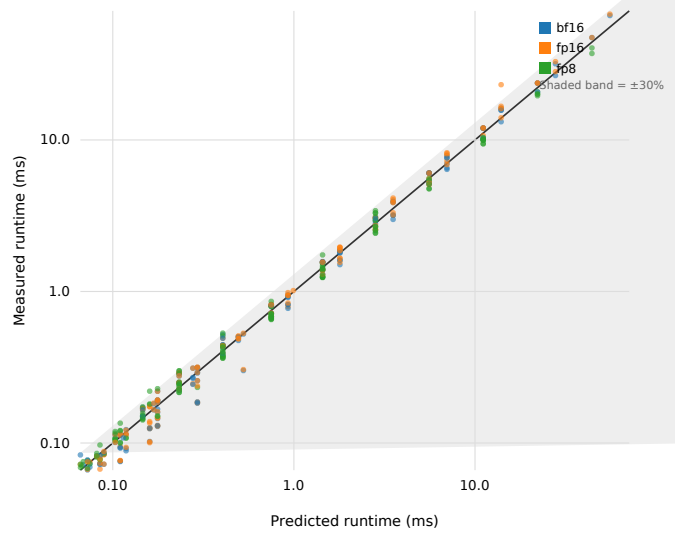


Fig. 4. Predicted versus measured runtime for the runtime-focused alternative predictor

Figure 4 shows predicted versus measured runtime for the refined model on all 435 validation and query configurations, with the identity line and 30% error bands. The points cluster around the identity line; the largest deviations are small-absolute-runtime configs where launch overhead and precision throughput assumptions interact.

All three supported precisions stay inside the success thresholds. FP8 is the best predicted; FP16 shows the largest

TABLE 3
Accuracy across passes of the runtime-focused alternative predictor.

Pass	Split	N	Base MAPE	Model MAPE	$\leq 30\%$	Bott. acc.
Emulator preliminary	validation	160	22.22	4.50	100.0	100.0
First real-hardware pass	validation	180	42.10	62.14	47.2	89.4
First real-hardware pass	query	504	42.30	65.20	41.1	85.9
Improved real-hardware pass	validation	105	55.64	12.62	93.3	60.0 (NCU)
Improved real-hardware pass	query	330	43.12	10.01	96.4	74.3 (NCU)
Refined real-hardware pass	validation	105	55.64	12.62	93.3	100.0 (NCU)
Refined real-hardware pass	query	330	43.12	10.01	96.4	100.0 (NCU)

TABLE 4
Final refined-model accuracy by precision for the runtime-focused alternative predictor.

Precision	Val. N	Val. MAPE	Val. $\leq 30\%$	Query N	Query MAPE	Query $\leq 30\%$
BF16	35	11.23	94.3	110	11.12	93.6
FP16	35	16.74	85.7	110	10.10	95.5
FP8	35	9.90	100.0	110	8.81	100.0

validation MAPE, driven by a few small-seqlen/high-head configurations where the measured runtime is small and the launch-overhead term dominates the percentage error.

The final model predicts MMA as the dominant bottleneck for most configurations, consistent with the NCU observation that every profiled config is compute-bound. Figure 5 shows the residual distribution by predicted bottleneck. The MMA residuals are centred near zero with a small positive bias for the very smallest configs.

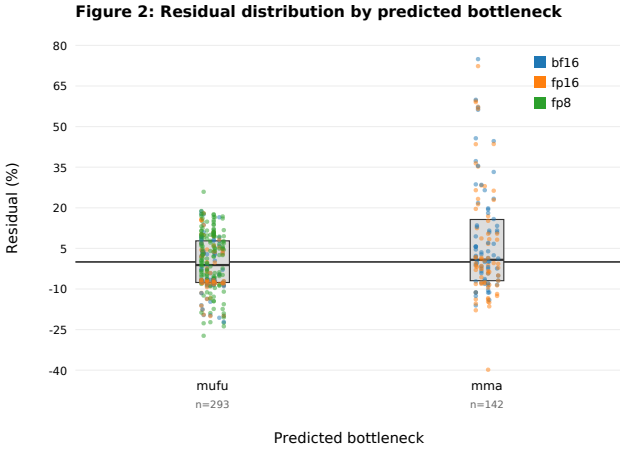


Fig. 5. Residual distribution by predicted bottleneck for the runtime-focused alternative predictor

The five largest validation errors are all small-sequence-length configurations with large batch and head counts, where the measured runtime is below 0.4 ms. The single largest validation APE is 74.9% (batch=8, heads=64, seqlen=1024, head_dim=128, causal=False, bf16). These cases are structural: when the absolute runtime is dominated by launch latency, small errors in the overhead estimate translate into large percentage errors. They do not represent calibration overfitting; the model is calibrated on a disjoint split.

Appendix B: Cycle-level hardware execution model

The roofline predictor in Appendix A collapses each memory domain into a single effective-bandwidth term. To make the

hardware path explicit, we built a cycle-level component model that follows one FA4 tile block through the B200 execution system (Figure 6). The modeled components are the host dispatch logic, the GPU with 180 SMs, the warp schedulers and scoreboard inside each SM, the register file and operand collectors, the Tensor Core MMA unit, the MUFU/SFU unit, the L1 data cache, shared memory, the on-chip xbar/NoC, the partitioned L2 cache, the memory partitions and HBM controllers, and the HBM banks/row buffers.

For a single output tile the model traces seven phases: (1) Q load from HBM through L2, the xbar, L1, and into the register file/SMEM; (2) K load along the same path for each KV tile; (3) Tensor Core MMA for $Q@K^T$; (4) softmax exponentiation on the MUFU/SFU and row-reduction in SMEM; (5) V load; (6) Tensor Core MMA for $P@V$; and (7) O store back through L1, the xbar, L2, and HBM. Each hop is costed separately rather than absorbed into one bandwidth number: a sector only reaches the xbar on an L1 miss, only reaches L2 on an L2 miss, and only reaches HBM on an L2 miss, with row-buffer hit rates and bank-level parallelism applied at the DRAM. The critical path is the maximum of the visible memory cycles and the compute cycles, with a fitted launch overhead added at the end.

This component-level view does not replace the roofline predictor used for the 540-config evaluation, but it explains why the improved predictor’s effective-bandwidth and launch-overhead corrections are physically necessary and provides the execution-system map shown in Figure 6.

The cycle-level model was calibrated on 16 B200 configurations and validated on 8 held-out configurations. It traces each data hop through the components shown in Figure 6 and predicts runtime from per-component cycle counts plus a launch-overhead term. On the validation split it achieves 13.16% MAPE, compared with 14.89% for the improved roofline predictor and 56.23% for the baseline roofline predictor on the same split. Calibration MAPE is 7.37%.

Appendix C: Sub-core partition scheduling model

Round 2 refines the per-SM reservation model by modelling each Blackwell SM as four independent sub-core partitions (Figure 7). Each partition has its own warp scheduler and can

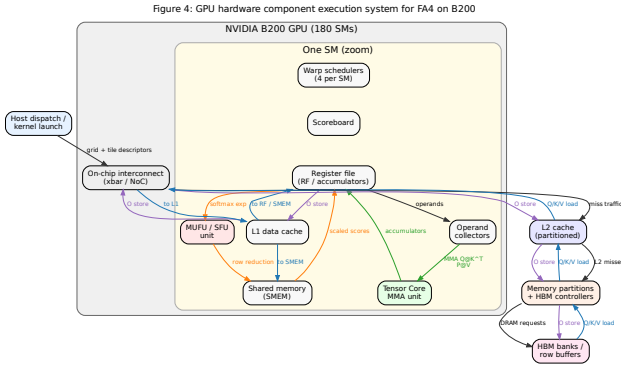


Fig. 6. GPU hardware component execution system for FlashAttention-4 on B200

issue one instruction per cycle to one execution unit. Warps are distributed across partitions, and the SM makespan is the busiest partition’s instruction schedule.

For each KV step the model counts warp-group MMA instructions, SFU `exp` instructions, FP32/INT32 scale/update instructions, and 32-byte memory transactions. These are distributed across the four partitions per SM. Per-partition cycles are

$$T_{\text{part}} = \frac{N_{\text{mma}}}{P \cdot S \cdot \eta_{\text{TC}}} + \frac{N_{\text{exp}}}{P \cdot S \cdot \eta_{\text{SFU}}} + \frac{N_{\text{update}}}{P \cdot S \cdot \eta_{\text{FP}}} + \frac{N_{\text{mem}}}{P \cdot S \cdot \eta_{\text{mem}}},$$

where $P = 4$ is the number of partitions and S is the number of active SMs. A calibrated partition-imbalance factor accounts for non-uniform instruction mixes. The global memory path and stage-overlap treatment are the same as in Section 3.2.

On the 16/8 split the sub-core partition model achieves 8.39% calibration MAPE and 16.41% validation MAPE (Table 2), slightly improving on the Round 1 reservation model while exposing the intra-SM scheduler bottleneck. The dominant bottleneck remains Tensor Core issue slots for large head-dim-128 tiles and the memory pipe for small or low-precision tiles.



Fig. 7. Sub-core partition scheduling inside one Blackwell SM